

Chapter 1: Introduction to Computer Security

1.1 The Security Problem in Computing

With the rapid growth of computers, the internet, cloud services, and mobile devices, information has become one of the most valuable assets for individuals, organizations, and governments. Almost every activity today—banking, education, healthcare, business operations, and communication—depends on computer systems.

However, this dependence has also introduced serious **security problems**.

The **security problem in computing** refers to the challenge of protecting computer systems, networks, and data from unauthorized access, misuse, damage, or disruption. As systems become more complex and interconnected, ensuring security becomes increasingly difficult.

Computer security is no longer limited to protecting a single machine. It involves securing:

- Data stored in computers
- Data transmitted over networks
- Applications and operating systems
- Hardware devices and infrastructure
- Human users who interact with systems

1.2 Meaning of Computer Security

Computer Security is the process of protecting computer systems and information from theft, damage, unauthorized access, and disruption.

Definition

Computer security can be defined as:

The protection of computer systems and data against unauthorized access, attacks, or destruction, while ensuring their availability and reliability.

Objectives of Computer Security

The primary objectives of computer security are:

1. **Protecting data from unauthorized access**
2. **Preventing data loss or corruption**
3. **Ensuring reliable system operation**
4. **Maintaining user trust**

In simple terms, computer security ensures that:

- Only authorized users can access the system
 - Data remains accurate and unchanged
 - Systems are available when needed
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The Vulnerability-Threat-Control Paradigm

The goal of computer security is protecting valuable assets. To study different ways of protection, we use a framework that describes how assets may be harmed and how to counter or mitigate that harm. A vulnerability is a weakness in the system, for example, in procedures, design, or implementation, that might be exploited to cause loss or harm. For instance, a particular system may be vulnerable to unauthorized data manipulation because the system does not verify a user's identity before allowing data access.

A vulnerability is a weakness that could be exploited to cause harm.

A threat to a computing system is a set of circumstances that has the potential to cause loss or harm. To see the difference between a threat and a vulnerability, consider the illustration in figure below. Here, a wall is holding water back. The water to the left of the wall is a threat to the

man on the right of the wall: The water could rise, overflowing onto the man, or it could stay beneath the height of the wall, causing the wall to collapse. So the threat of harm is the potential for the man to get wet, get hurt, or be drowned. For now, the wall is intact, so the threat to the man is unrealized.

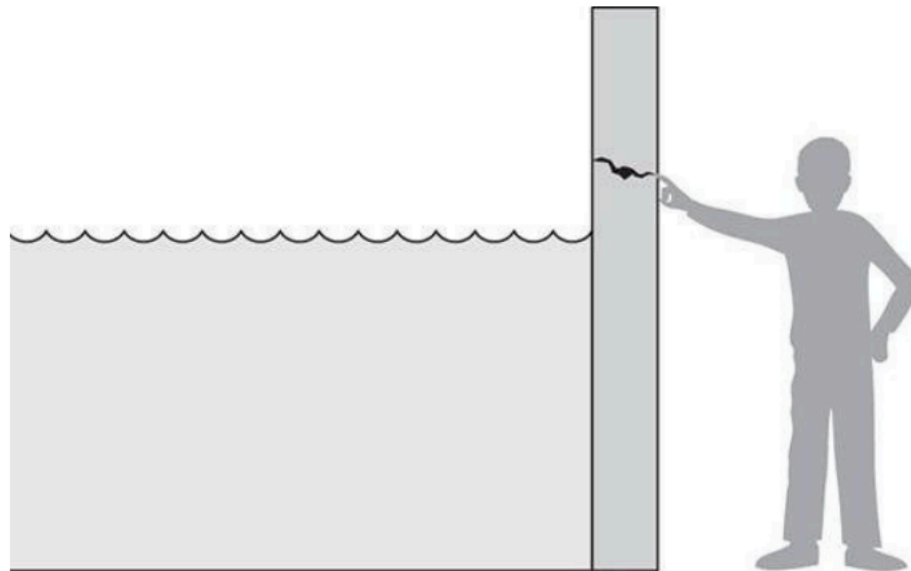


FIGURE 1-4 Threat and Vulnerability

A threat is a set of circumstances that could cause harm.

However, we can see a small crack in the wall—a vulnerability that threatens the man’s security. If the water rises to or beyond the level of the crack, it will exploit the vulnerability and harm the man. There are many threats to a computer system, including human-initiated and computerinitiated ones. We have all experienced the results of inadvertent human errors, hardware design flaws, and software failures. But natural disasters are threats, too; they can bring a system down when the computer room is flooded or the data center collapses from an earthquake, for example. A human who exploits a vulnerability perpetrates an attack on the system. An attack can also be launched by another system, as when one system sends an overwhelming flood of messages to another, virtually shutting down the second system’s ability to function. Unfortunately, we have seen this type of attack frequently, as denial-of-service attacks deluge servers with more messages than they can handle.

1.3 Computer Criminals

A **computer criminal** (also known as a cybercriminal) is a person or group that uses computers, networks, or digital technologies to commit illegal or unethical activities.

Types of Computer Criminals

1. Hackers / Crackers

Hackers are individuals who gain unauthorized access to computer systems.

They can be categorized as:

- **White Hat Hackers:** Ethical hackers who help improve security
- **Black Hat Hackers:** Malicious hackers who exploit systems
- **Grey Hat Hackers:** Operate between ethical and unethical boundaries

2. Script Kiddies

These are inexperienced attackers who use pre-written tools or scripts without understanding how they work.

3. Insider Threats

Employees or authorized users who misuse their access intentionally or unintentionally.

4. Cyber Terrorists

Individuals or groups who attack systems to create fear, disrupt services, or achieve political or ideological goals.

1.4 Methods of Defense

To counter security threats, various **methods of defense** are used in computer systems.

1. Authentication

Authentication verifies the identity of users.

Examples:

- Username and password
- Biometric systems (fingerprint, face recognition)
- One-Time Passwords (OTP)

2. Authorization

Authorization determines what an authenticated user is allowed to do.

3. Encryption

Encryption converts data into unreadable form to prevent unauthorized access.

It is widely used in:

- Online banking
- Secure emails
- Data storage

4. Firewalls

A firewall monitors and controls incoming and outgoing network traffic based on security rules.

5. Antivirus and Anti-malware Software

These tools detect and remove malicious software such as viruses, worms, and spyware.

6. Regular Updates and Patching

Software updates fix security vulnerabilities and bugs.

1.5 Elements of Computer Security

Computer security is built around three fundamental elements, commonly known as the **CIA Triad**.

1. Confidentiality

Confidentiality ensures that information is accessible only to authorized users.

Example:

- Password-protected files
- Encrypted communication

2. Integrity

Integrity ensures that data is accurate and has not been altered without authorization.

Example:

- Checksums
- Digital signatures

3. Availability

Availability ensures that systems and data are accessible when needed.

Example:

- Backup systems
- Fault-tolerant hardware

Together, these elements form the foundation of all security policies and mechanisms.

1.6 Threats in Information Technology

A **threat** is any potential danger that can exploit a vulnerability and cause harm to a system.

Common IT Threats

1. Malware

Malware includes viruses, worms, Trojan horses, ransomware, and spyware.

2. Phishing

Phishing attacks trick users into revealing sensitive information such as passwords and credit card details.

3. Denial of Service (DoS) Attacks

DoS attacks attempt to make systems unavailable by overwhelming them with traffic.

4. Data Breaches

Unauthorized access to confidential data, often resulting in data theft.

5. Social Engineering

Manipulating people into giving up confidential information.

6. Physical Threats

Fire, theft, power failure, and natural disasters can also threaten computer systems.

1.7 Information Security vs Network Security

Although closely related, **Information Security** and **Network Security** are not the same.

Information Security

Information Security focuses on protecting data in all forms:

- Digital data
- Printed documents
- Verbal communication

Key focus areas:

- Data confidentiality
- Data integrity
- Data availability

Network Security

Network Security focuses specifically on protecting network infrastructure and data during transmission.

Key focus areas:

- Firewalls
- Intrusion Detection Systems (IDS)
- Secure communication protocols

Differences Between Information Security and Network Security

Aspect	Information Security	Network Security
Scope	Protects all information	Protects network data
Focus	Data protection	Network protection
Covers	Policies, procedures, people, technology	Routers, switches, firewalls
Dependency	Independent of networks	Depends on networks